

Cloud Computing

BRANCH: INFORMATION TECH (IT)

COURSE CODE: IT302

Subject Overview & Scope

This comprehensive document serves as the official compiled study notes for Cloud Computing (IT302) under the Information Tech (IT) department. It is structured into four core university syllabus units covering primary concepts, definitions, formulas, and advanced technical applications for university examinations and placements.

University Course Syllabus & Core Units

Unit 1: Introduction to Cloud

NIST definition of cloud computing, benefits, service models (IaaS, PaaS, SaaS), deployment models (Public, Private, Hybrid).

Unit 2: Virtualization Technology

Hypervisors: Type 1 and Type 2. Virtual machines, CPU virtualization, memory virtualization. Containers vs Virtual Machines (Docker).

Unit 3: Cloud Storage & Services

Object storage (AWS S3) vs Block storage (EBS). Cloud database services. Serverless computing: AWS Lambda, Azure Functions.

Unit 4: Security & DevOps in Cloud

Identity and Access Management (IAM), encryption, firewalls. DevOps pipeline: CI/CD in cloud, infrastructure as code (Terraform).

Availability = $MTBF / (MTBF + MTTR)$. IaaS (Infrastructure), PaaS (Platform), SaaS (Software) definitions.

CRITICAL FORMULA / PRINCIPLE:

Key Formulas & Exam Pointers



